Homework #0

CSEP 546: Machine Learning
Professor Jamie Morgenstern
Due: ~~**Wednesday April 8, 2026 11:59pm**~~ **Updated: Thursday April 9, 2026 6:25pm (just before class)**
38 points

Probability and Statistics

A1. *[2 points]* (From Murphy Exercise 2.4.) After your yearly checkup, the doctor has bad news and good news. The bad news is that you tested positive for a serious disease, and that the test is 99% accurate (i.e., the probability of testing positive given that you have the disease is 0.99, as is the probability of testing negative given that you don't have the disease). The good news is that this is a rare disease, striking only one in 10,000 people. What are the chances that you actually have the disease?

What to Submit:

- Final Answer
- Corresponding Calculations

Solution:

Known Information: The disease is extremely rare (1 in 10,000), so a 99% accurate test will produce ∼101 positive tests out of 10,000 people, of which only 1 person actually has the disease—meaning 1 true positive and 100 false positives.
Therefore, the total probability of testing positive comes from two groups: people with the disease who test positive + people without the disease who test positive.

Definitions:

- Hypothesis H: you have the disease
- Prior: $P(H) = \frac{1}{10{,}000} = 0.0001$ (rate of the disease in the population)
- $P(E \mid H) = 0.99$ (true positive: test is positive given you have the disease)
- $P(E \mid \neg H) = 0.01$ (false positive: test is positive given you don't have the disease)

Step 1: Compute $P(E)$ — the probability of testing positive across the entire population.
This is the sum of:

- People with disease testing positive: $P(E \mid H) \cdot P(H)$
- People with no disease testing positive: $P(E \mid \neg H) \cdot P(\neg H)$

$$\begin{aligned} P(E) &= P(E \mid H) \cdot P(H) + P(E \mid \neg H) \cdot P(\neg H) \\ &= (0.99)(0.0001) + (0.01)(0.9999) \\ &= 0.000099 + 0.009999 \\ &= 0.010098 \end{aligned}$$

Step 2: Apply Bayes' Theorem — of all the people who tested positive, what fraction truly have the disease.

$$\begin{aligned} P(H \mid E) &= \frac{P(E \mid H) \cdot P(H)}{P(E)} \\ &= \frac{(0.99)(0.0001)}{0.010098} \\ &= \frac{0.000099}{0.010098} \\ &= 0.00980392156 \end{aligned}$$

Final Answer: $P(H \mid E) \approx 0.9804\%$

Collaboration: `hw0/chat_conversation_export_A1.md` in Appendix section.

A2. Let $\mathbf{X} = \begin{bmatrix} X_1 & \cdots & X_n \end{bmatrix}^\top$ be a random vector with mean vector $\boldsymbol{\mu} \in \mathbb{R}^n$ and covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{n \times n}$. That is, for each $i \in [n]$, $\mathbb{E}[X_i] = \boldsymbol{\mu}_i$, and for each $i, j \in [n]$, $\mathrm{Cov}(X_i, X_j) = \boldsymbol{\Sigma}_{i,j}$.

a. *[2 points]* Let $i \in [n]$. Find an expression for $\mathbb{E}\left[X_i^2\right]$ in terms of only $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$.

b. *[2 points]* Recall that the trace of a square matrix is the sum of its diagonal entries. Show that $\mathbb{E}\left[\mathbf{X}^\top \mathbf{X}\right] = \mathrm{tr}(\boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \boldsymbol{\mu}$.

What to Submit:

- **Part a:** Expression for $\mathbb{E}\left[X_i^2\right]$ and corresponding calculations
- **Part b:** Proof

Solution:

Part a: Find an expression for $\mathbb{E}\left[X_i^2\right]$

Known Information:

- $\mathbf{X} = [X_1, \ldots, X_n]^\top$ is a random vector
- $\mathbb{E}[X_i] = \boldsymbol{\mu}_i$ (mean of the i-th dimension/feature)
- $\mathrm{Cov}(X_i, X_j) = \boldsymbol{\Sigma}_{i,j}$ (covariance matrix entry)
- From this we can deduce: $\mathrm{Var}[X_i] = \boldsymbol{\Sigma}_{i,i}$ (diagonal entries = variances, i.e. $i = j$)

Definitions:

- Variance is the expected value of the squared deviation from the mean $\boldsymbol{\mu}$ of a random variable:

$$\mathrm{Var}[X] = \mathbb{E}\left[(X - \mu)^2\right]$$

- When c is a scalar/constant: $\mathbb{E}[c] = c$ and $\mathbb{E}[cX] = c\,\mathbb{E}[X]$
- And $\mu = \mathbb{E}[X]$

Step 1: Relation between μ and variance.

$$\begin{aligned}\mathrm{Var}[X] &= \mathbb{E}\big[(X-\mu)^2\big] \\ &= \mathbb{E}\big[X^2 - 2X\mu + \mu^2\big] \\ &= \mathbb{E}\big[X^2\big] - \mathbb{E}[2X\mu] + \mathbb{E}\big[\mu^2\big]\end{aligned}$$

Step 2: 2 and μ are constants and $\mathbb{E}[c] = c$; replace constants and simplify.

$$\begin{aligned}\mathrm{Var}[X] &= \mathbb{E}\big[X^2\big] - 2\mu\ \mathbb{E}[X] + \mu^2 \\ &= \mathbb{E}\big[X^2\big] - 2\mu^2 + \mu^2 \\ &= \mathbb{E}\big[X^2\big] - \mu^2\end{aligned}$$

Therefore:

$$\begin{aligned}\boldsymbol{\Sigma}_{i,i} &= \mathbb{E}\big[X_i^2\big] - \boldsymbol{\mu}_i^2 \\ \mathbb{E}\big[X_i^2\big] &= \boldsymbol{\Sigma}_{i,i} + \boldsymbol{\mu}_i^2\end{aligned}$$

Final Answer: $\boxed{\mathbb{E}\big[X_i^2\big] = \boldsymbol{\Sigma}_{i,i} + \boldsymbol{\mu}_i^2}$

Part b: Show that $\mathbb{E}\big[\mathbf{X}^\top\mathbf{X}\big] = \mathrm{tr}(\boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top\boldsymbol{\mu}$

$\mathbf{X} = [X_1, \ldots, X_n]^\top$ is a random vector, so its dot product is:

$$\mathbf{X}^\top\mathbf{X} = X_1^2 + X_2^2 + X_3^2 + \cdots + X_n^2$$

Expectation on the right-hand side can be written as a sum of individual expectations:

$$\mathbb{E}\big[\mathbf{X}^\top\mathbf{X}\big] = \mathbb{E}\big[X_1^2\big] + \mathbb{E}\big[X_2^2\big] + \mathbb{E}\big[X_3^2\big] + \cdots + \mathbb{E}\big[X_n^2\big]$$

Now replace using the relation derived in Part a:

$$\begin{aligned}\mathbb{E}\big[\mathbf{X}^\top\mathbf{X}\big] &= (\boldsymbol{\Sigma}_{1,1} + \boldsymbol{\mu}_1^2) + (\boldsymbol{\Sigma}_{2,2} + \boldsymbol{\mu}_2^2) + (\boldsymbol{\Sigma}_{3,3} + \boldsymbol{\mu}_3^2) + \cdots + (\boldsymbol{\Sigma}_{n,n} + \boldsymbol{\mu}_n^2) \\ &= (\boldsymbol{\Sigma}_{1,1} + \boldsymbol{\Sigma}_{2,2} + \boldsymbol{\Sigma}_{3,3} + \cdots + \boldsymbol{\Sigma}_{n,n}) + (\boldsymbol{\mu}_1^2 + \boldsymbol{\mu}_2^2 + \boldsymbol{\mu}_3^2 + \cdots + \boldsymbol{\mu}_n^2)\end{aligned}$$

- $(\boldsymbol{\Sigma}_{1,1} + \boldsymbol{\Sigma}_{2,2} + \boldsymbol{\Sigma}_{3,3} + \cdots + \boldsymbol{\Sigma}_{n,n})$ is the trace of the covariance matrix $= \mathrm{tr}(\boldsymbol{\Sigma})$
- $(\boldsymbol{\mu}_1^2 + \boldsymbol{\mu}_2^2 + \boldsymbol{\mu}_3^2 + \cdots + \boldsymbol{\mu}_n^2)$ is the dot product $= \boldsymbol{\mu}^\top\boldsymbol{\mu}$

Final Answer: $\boxed{\mathbb{E}\big[\mathbf{X}^\top\mathbf{X}\big] = \mathrm{tr}(\boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top\boldsymbol{\mu}}$ □

Collaboration: `hw0/covariance_matrix_conversation_A2.md` in Appendix section.

A3. Let $X_1, X_2, ..., X_n \sim \mathcal{N}(\mu, \sigma^2)$ be i.i.d. random variables. Compute the following:

a. *[1 point]* $a \in \mathbb{R}, b \in \mathbb{R}$ such that $aX_1 + b \sim \mathcal{N}(0, 1)$.

b. *[2 points]* Using a and b from the previous part, $\mathbb{E}[|aX_1 + b|]$.
Hint: You may need to use u-substitution.

c. *[1 point]* $\mathbb{E}[X_1 + 2X_2], \mathrm{Var}[X_1 + 2X_2]$.

d. *[2 points]* Setting $\widehat{\mu}_n = \frac{1}{n}\sum_{i=1}^n X_i$, the mean and variance of $\sqrt{n}(\widehat{\mu}_n - \mu)$.

What to Submit:

- **Part a:** a, b
- **Part b:** $\mathbb{E}[|aX_1 + b|]$
- **Part c:** $\mathbb{E}[X_1 + 2X_2]$, $\mathrm{Var}[X_1 + 2X_2]$
- **Part d:** $\mathbb{E}[\sqrt{n}(\widehat{\mu}_n - \mu)]$, $\mathrm{Var}[\sqrt{n}(\widehat{\mu}_n - \mu)]$
- **Parts a-d** Corresponding calculations

Linear Algebra and Vector Calculus

A4. Let $\mathbf{A} = \begin{bmatrix} 0 & 1 & -1 & -2 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 2 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} -1 & 2 & 0 \\ -2 & 1 & 2 \\ 2 & 0 & -3 \\ -1 & -1 & 2 \end{bmatrix}$.

a. *[1 point]* Compute $\mathbf{AB}$.

b. *[2 points]* What is the rank of $\mathbf{AB}$? Use this to determine the rank of both $\mathbf{A}$ and $\mathbf{B}$.

c. *[1 point]* Find a matrix $\mathbf{C}$ such that $\mathbf{ABC} = \mathbf{I}$, or explain why such a matrix does not exist.

d. *[1 point]* Let $b = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^\top$. What is the solution to the linear system $\mathbf{AB}x = b$?

What to Submit:

- **Parts a-d:** Solution and corresponding calculations

A5. For possibly non-symmetric $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ and $c \in \mathbb{R}$, let $f(x, y) = x^\top \mathbf{A} x + y^\top \mathbf{B} x + c$. Define

$$\nabla_z f(x, y) = \begin{bmatrix} \frac{\partial f}{\partial z_1}(x, y) & \frac{\partial f}{\partial z_2}(x, y) & \dots & \frac{\partial f}{\partial z_n}(x, y) \end{bmatrix}^\top \in \mathbb{R}^n .$$

Note: If you are unfamiliar with gradients, you may find the resources available on the course website useful. Section 4 of Zico Kolter and Chuong Do's Linear Algebra Review and Reference may be particularly helpful.

a. *[2 points]* Explicitly write out the function $f(x, y)$ in terms of the components $A_{i,j}$ and $B_{i,j}$ using appropriate summations over the indices.

b. *[2 points]* What is $\nabla_x f(x, y)$ and $\nabla_y f(x, y)$ in terms of summations over indices?

c. *[2 points]* What is $\nabla_x f(x, y)$ and $\nabla_y f(x, y)$ in vector notation?

What to Submit:

- **Part a:** Explicit formula for $f(x, y)$
- **Parts b:** Summation form and corresponding calculations. Summation form includes writing out what each component of the resultant vector is, where each component is expressed as a summation. Intermediate components may be indicated by ellipses, like in the equation given in the problem description.
- **Part c:** Vector form and corresponding calculations. Vector form includes writing the final answer only in terms of products, sums (or differences), and/or transposes of the input matrices and vectors.

A6. Show the following:

a. *[2 points]* Let $g\colon \mathbb{R}\setminus\{0\} \to \mathbb{R}$ and $v, w \in \mathbb{R}^n$ such that $v_i \neq 0$ and $g(v_i) = w_i$ for $i \in [n]$. Find an expression for g such that $\text{diag}(v)^{-1} = \text{diag}(w)$.

b. *[2 points]* Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be orthonormal and $x \in \mathbb{R}^n$. An orthonormal matrix is a square matrix whose columns and rows are orthonormal vectors, such that $\mathbf{A}\mathbf{A}^\top = \mathbf{A}^\top\mathbf{A} = \mathbf{I}$ where $\mathbf{I}$ is the identity matrix. Show that $||\mathbf{A}x||_2^2 = ||x||_2^2$.

c. *[2 points]* Let $\mathbf{B} \in \mathbb{R}^{n\times n}$ be invertible and symmetric. A symmetric matrix is a square matrix satisfying $\mathbf{B} = \mathbf{B}^\top$. Show that $\mathbf{B}^{-1}$ is also symmetric.

d. *[2 points]* Let $\mathbf{C} \in \mathbb{R}^{n\times n}$ be positive semi-definite (PSD). A positive semi-definite matrix is a symmetric matrix satisfying $x^\top\mathbf{C}x \geq 0$ for any vector $x \in \mathbb{R}^n$. Show that its eigenvalues are non-negative.

What to Submit:

- **Part a:** Explicit formula for g
- **Parts a-d:** Proof

Programming

These problems are available in a .zip file, with some starter code. All coding questions in this class will have starter code. **Before attempting these problems, you will need to set up a uv environment that you will use for every assignment in the course. Unzip the HW0-A.zip file and read the instructions in the README file to get started.**

A7. For $\nabla_x f(x, y)$ as solved for in Problem 5:

a. *[1 point]* Using native Python, implement `vanilla_solution` using your `vanilla_matmul` and `vanilla_transpose` functions.

b. *[1 point]* Now implement `numpy_version` using NumPy functions.

c. *[1 point]* Report the difference in wall-clock time for parts a-b, and discuss reasons for the observed difference.

What to Submit:

- **Part c:** Plot that shows the difference in wall-clock time for parts a-b
- **Part c:** Explanation for the difference (1-2 sentences)
- **Code** on Gradescope through coding submission

A8. Two random variables X and Y have equal distributions if their CDFs, F_X and F_Y, respectively, are equal, i.e. for all x, $|F_X(x) - F_Y(x)| = 0$. The central limit theorem says that the sum of k independent, zero-mean, variance $1/k$ random variables converges to a (standard) Normal distribution as k tends to infinity. We will study this phenomenon empirically (you will use the Python packages NumPy and Matplotlib). Each of the following subproblems includes a description of how the plots were generated; these have been coded for you. The code is available in the .zip file. In this problem, you will add to our implementation to explore **matplotlib** library, and how the solution depends on n and k.

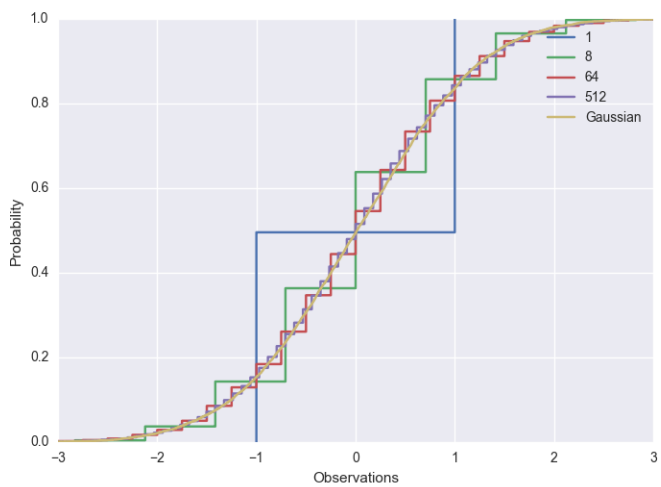
a. *[2 points]* For $i = 1, \ldots, n$ let $Z_i \sim \mathcal{N}(0,1)$. Let $x \mapsto F(x)$ denote the true CDF from which each Z_i is drawn (i.e., Gaussian). Define $\widehat{F}_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{Z_i \leq x\}$ for $x \in \mathbb{R}$ and we will choose n large enough such that, for all $x \in \mathbb{R}$,

$$\sqrt{\mathbb{E}\left[\left(\widehat{F}_n(x) - F(x)\right)^2\right]} \leq 0.0025 \ .$$

Plot $x \mapsto \widehat{F}_n(x)$ for x ranging from -3 to 3.

b. *[2 points]* Define $Y^{(k)} = \frac{1}{\sqrt{k}}\sum_{i=1}^{k} B_i$ where each B_i is equal to -1 and 1 with equal probability and the B_i's are independent. We know that each $\frac{1}{\sqrt{k}}B_i$ is zero-mean and has variance $1/k$. For each $k \in \{1, 8, 64, 512\}$ we will generate n (same as in part a) independent copies $Y^{(k)}$ and plot their empirical CDF on the same plot as part a.

Be sure to always label your axes. Your plot should look something like the following (up to styling) (Tip: checkout `seaborn` for instantly better looking plots.)



What to Submit:

- **Part a:** Value for n (You can simply print the value determined by the code provided for you). **Part b:** In 1-2 sentences: How does the empirical CDF change with k?
- **Parts a and b:** Plot of $x \mapsto \widehat{F}_n(x)$ for $x \in [-3, 3]$
- **Code** on Gradescope through coding submission

Appendix: AI Collaboration Transcripts

A1 — Bayes' Theorem (chat_conversation_export_A1.md)

User:
After your yearly checkup, the doctor has bad news and good news. The bad news is that you tested positive for a serious disease, and that the test is 99% accurate (i.e., the probability of testing positive given that you have the disease is 0.99, as is the probability of testing negative given that you don't have the disease). The good news is that this is a rare disease, striking only one in 10,000 people. What are the chances that you actually have the disease?
Please include links only don't describe the solution or the details of the concepts. I want to solve the problem after reading the material, don't solve the problem.

Assistant:

- Khan Academy — Tree diagrams and conditional probability
- MIT OpenCourseWare — Conditional Probability, Independence, Bayes' Theorem (18.05)
- MIT OpenCourseWare — Conditional Probability Lecture Notes (Mathematics for Computer Science)
- Seeing Theory — Bayesian Inference
- Wikipedia — Conditional probability
- Wikipedia — Base rate fallacy

User:
My intuition, look at false positive and true positives of a test in population and use that information to find out probability. Is that right direction? Don't solve the problem.

Assistant:
Yes. That is the right direction.
You want to think in terms of:

- how many people in a large population actually have the disease
- how many of those test positive
- how many people do **not** have the disease
- how many of those still test positive anyway

Then compare:

- **true positives**

against

- **all positives**

That framing is exactly the useful one here. Another good lens to review is **conditional probability / Bayes' theorem**, but your population-count intuition is often the cleanest way to reason it out.

User:
Download this chat conversation as an MD file.

A2 — Covariance Matrix (covariance_matrix_conversation_A2.md)

User:
Explain covariance matrix with 3 components/dimensions with a real world example.

Assistant:
Think of a covariance matrix as a compact way to describe how several quantities vary on their own and how they move together.
With **3 components**, the covariance matrix is a **3×3 table**.
Let's use a real-world example:
Suppose you track, over many days, these 3 things for a delivery company:

- X_1: number of packages delivered
- X_2: fuel used
- X_3: driver overtime hours

Then the covariance matrix is:

$$\mathbf{\Sigma} = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \mathrm{Cov}(X_1, X_3) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \mathrm{Cov}(X_2, X_3) \\ \mathrm{Cov}(X_3, X_1) & \mathrm{Cov}(X_3, X_2) & \mathrm{Var}(X_3) \end{bmatrix}$$

What each entry means:
Diagonal entries = variances — these tell you how much each variable fluctuates by itself.

- $\mathrm{Var}(X_1)$: how much package count changes from day to day
- $\mathrm{Var}(X_2)$: how much fuel usage changes
- $\mathrm{Var}(X_3)$: how much overtime changes

Big variance means that quantity is less stable.
Off-diagonal entries = covariances — these tell you whether two variables tend to rise and fall together.

- $\mathrm{Cov}(X_1, X_2)$: when deliveries go up, does fuel use also go up?
- $\mathrm{Cov}(X_1, X_3)$: when deliveries go up, does overtime also increase?
- $\mathrm{Cov}(X_2, X_3)$: when fuel use is high, is overtime usually high too?

Interpretation:

- **Positive covariance**: they tend to increase together
- **Negative covariance**: when one goes up, the other tends to go down
- **Near zero covariance**: no clear linear relationship

Example with numbers:
Imagine the covariance matrix from your data is:

$$\mathbf{\Sigma} = \begin{bmatrix} 100 & 40 & 25 \\ 40 & 64 & 20 \\ 25 & 20 & 16 \end{bmatrix}$$

Here is how to read it:

- **100**: package deliveries vary quite a bit
- **64**: fuel use also varies, but a little less

- **16**: overtime varies the least

And the relationships:

- **40** between deliveries and fuel: busier days usually use more fuel
- **25** between deliveries and overtime: more deliveries usually mean more overtime
- **20** between fuel and overtime: when routes are heavy enough to burn more fuel, overtime also tends to rise

Why it is a matrix:
Because with 3 variables, you need to describe each variable's own spread and each pairwise relationship. For 3 variables, that means 3 variances and 3 pairwise covariances. The matrix keeps all of that organized in one object.
Geometric intuition:
If you imagine each day as a point in 3D space (packages, fuel, overtime), then the covariance matrix tells you the **shape of the cloud of points**.

- Large variance in one direction → cloud spreads more along that axis
- Positive covariance between two variables → cloud tilts along those two axes
- Negative covariance → cloud tilts the opposite way

So the covariance matrix tells you both **how wide** the data is in each direction and **how tilted** the data is because variables move together.

User:
Export this conversation as an MD file.